10 MW Solar PV Power Plant

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10 MW Solar PV Power Plant

26/12/25 - Friday

- Project Description
- Objectives Involved
- Success Factors
- Appendix

Project Description

By installing and successfully operating 10 MW photovoltaic (PV) power plants will deliver electricity for consumption by the owners, the relevant peoples in the project assessment place will be made aware of the technical and economic potential of solar power generation. Furthermore, the power required from the public grid will be reduced, and overall expenditure on electric power will be lowered & our project aims to create the necessary awareness among the population, and especially among policymakers and large investors,Youngsters.....

Objectives Involved

Overall project objective:

Our project will make a contribution toward sustainable energy supply, and will serve as a showpiece demonstrating the potential for stable and strongly desiring power supply based on low-carbon energy production.

- Schedule objectives
- Permission objectives
- Financial objectives
- Technical objectives
- Special objectives

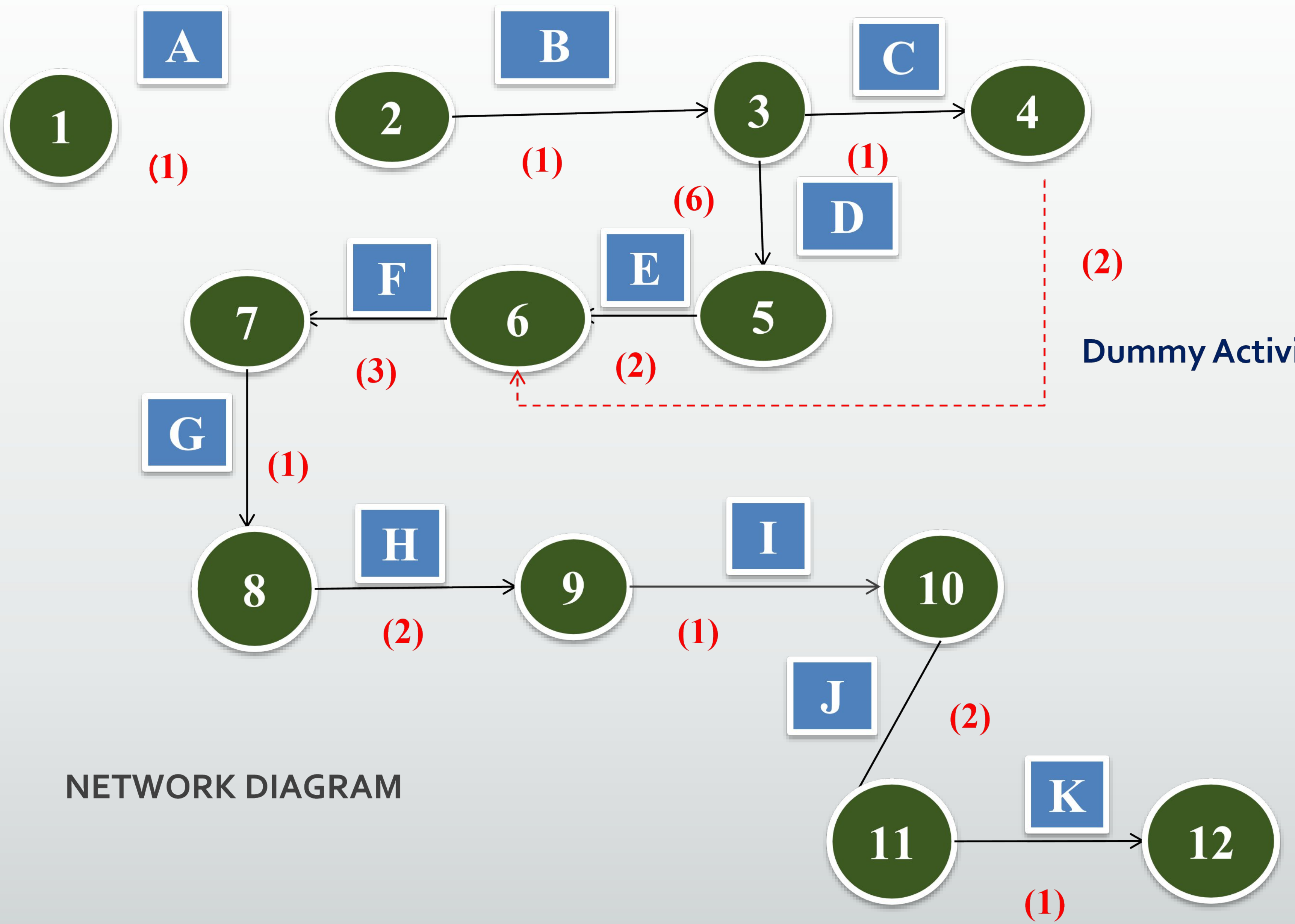
1.Schedule objectives

Scheduling

- Total Project Time 5 Months   Date of Ordering   2 Months
- Financial Closure Achievement 2 Months
- Plant & Machinery Ordering   1 Month
- DOO & FCA - During which detailed engineering, procurement, erection and commissioning of civil & structural, mechanical and electrical equipment will be executed.
- The plant & machinery ordering process will happen and the entire ordering process will happen in one month. During this period, all detailed engineering designs will be made ready for the execution phase to follow.

ACTIVITY	DESCRIPTION	PRECEEDING ACTIVITY	DURATION (WEEKS)
A	SITE ASSESSMENT	-	1
B	DESIGNING	A	1
C	LABOUR	B	1
D	PROCUREMENT	B	6
E	MOUNTING STRUCTURE (ERECTION)	C, D	2
F	PANEL ERECTION	E	3
G	JUNCTION BOX	F	1
H	INVERTER	G	2
I	CABLING	G	1

J	TRANSFORMER	H, I	2
K	COMMISSIONING	J	1



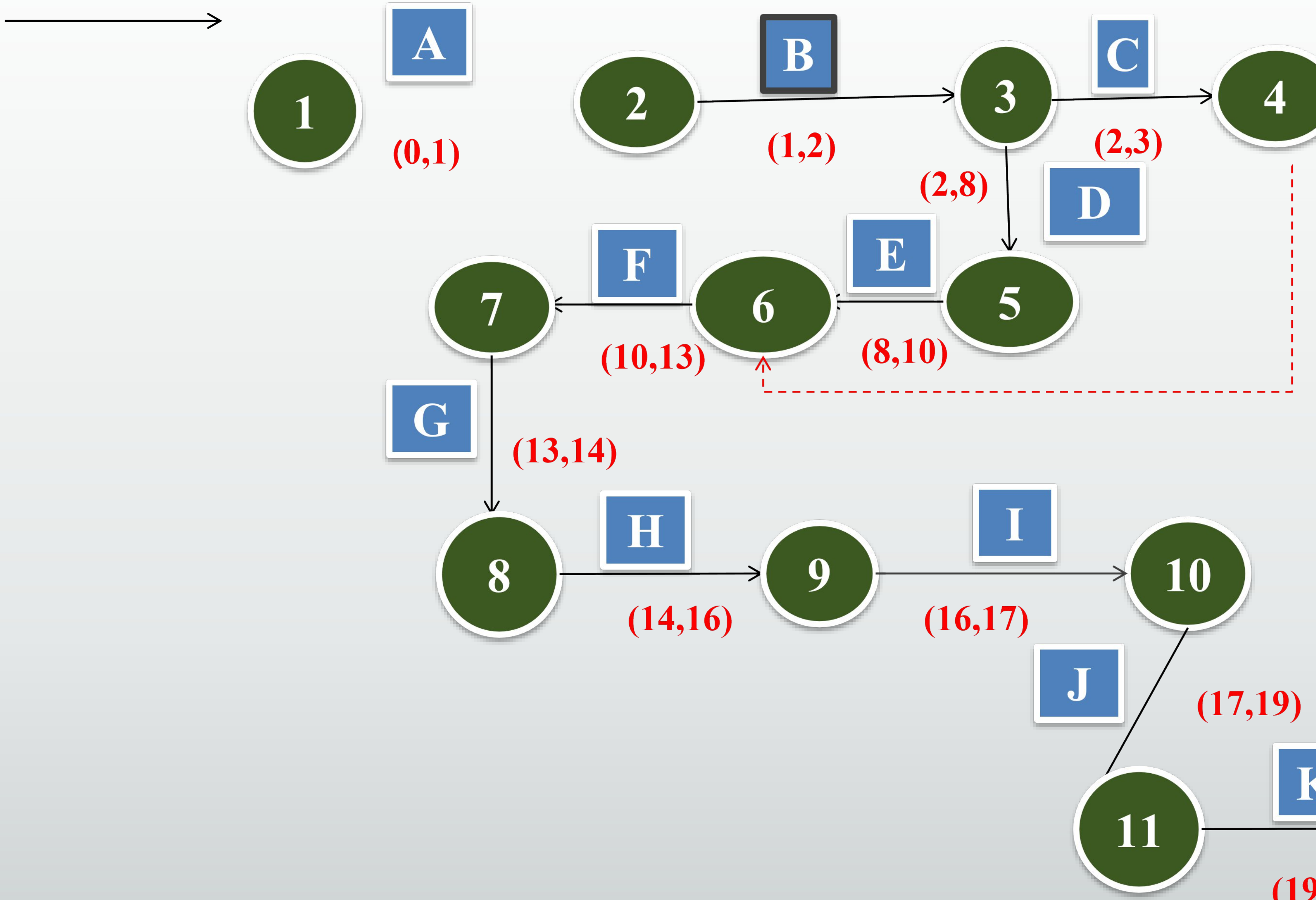
NETWORK DIAGRAM

Possible Pathways

A-B-C-E-F-G-H-J-K = 14 weeks

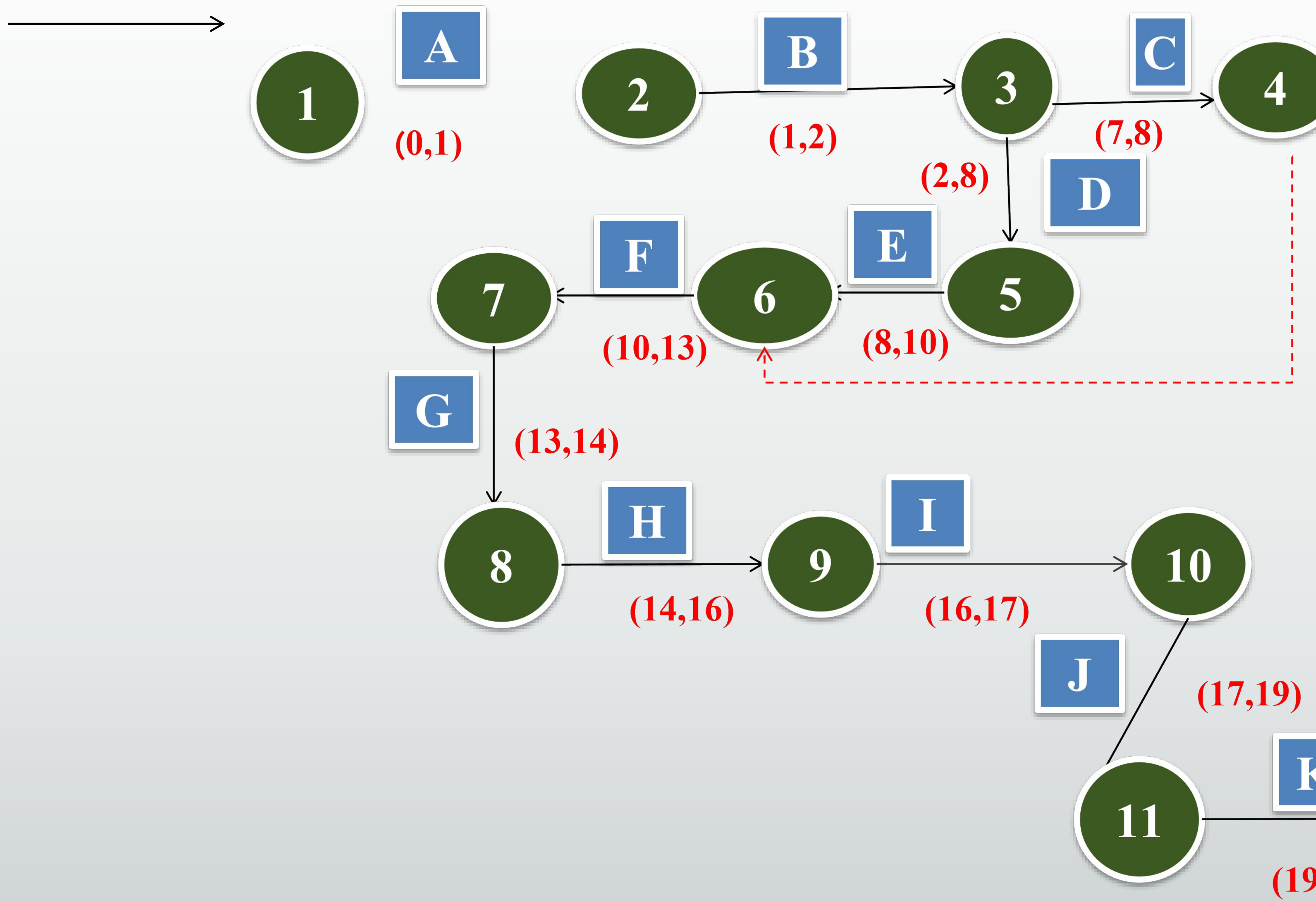
A-B-D-E-F-G-H-I-J-K =20 weeks

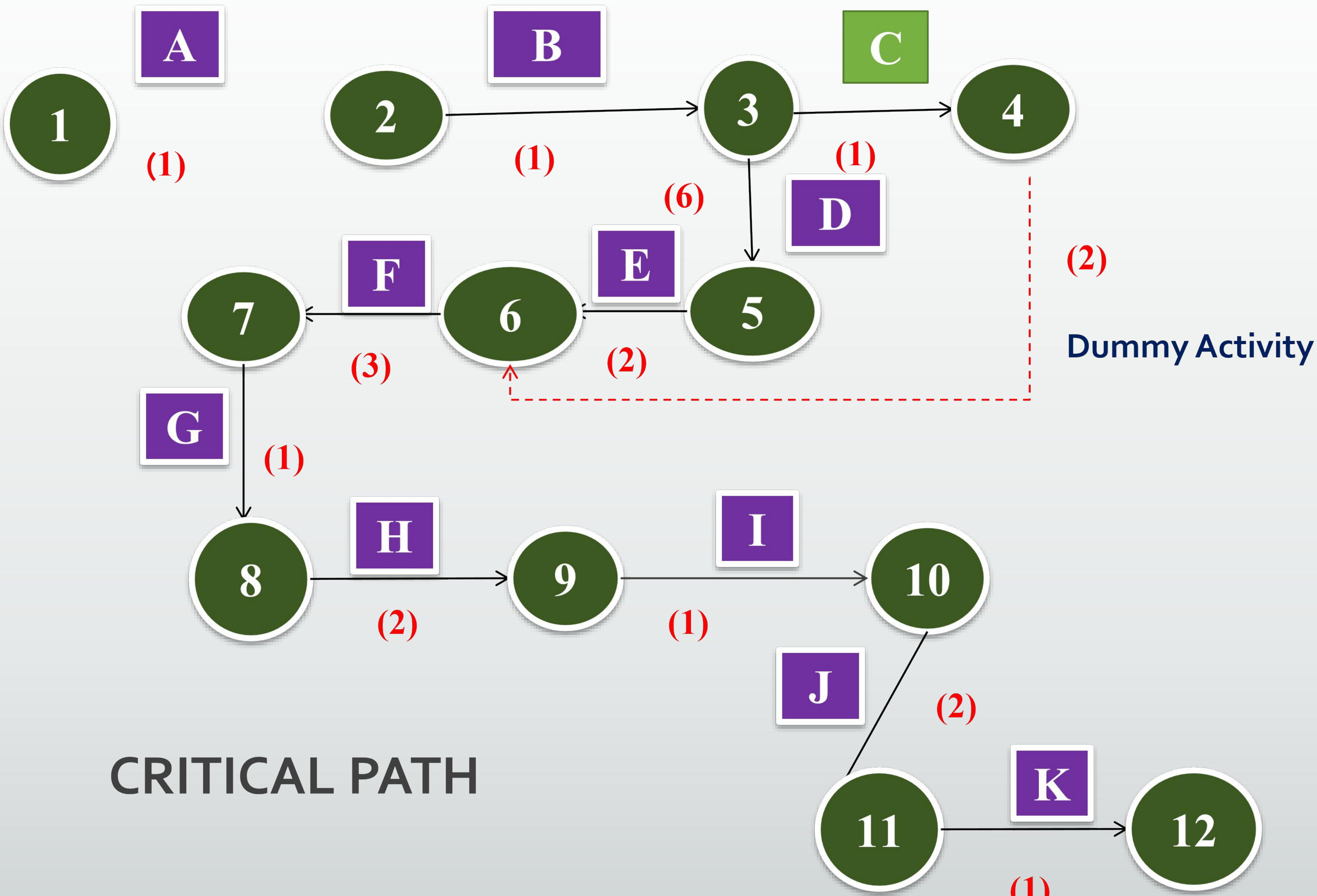
PERT- FORWARD PASS METHOD



1
A
(0,1)
2
B
3
(1,2)
C
4
(2,3)
(2,8)
D
E
5
F
6
7
(10,13)
(8,10)
G
(13,14)
H
8
9
I
10
(14,16)
(16,17)
J
(17,19)
11

PERT- BACKWARD PASS METHOD


1
A
(0,1)
2
B
(1,2)
3
C
4
(7,8)
(2,8)
D
E
F
7
6
5
(10,13)
(8,10)
G
(13,14)
8
H
9
I
10
(14,16)
(16,17)
J
(17,19)
11



CRITICAL PATH

2.Permission objectives

Permissions/Clearances

Accredited by State Load Dispatch Centre (SLDC) for REC mechanism


Industrial Clearance
Land conversion
EC Certificate from Gov.
Latest tax receipt from the Gov.
Fire Safety certificate from Fire Department
Contract labour license from Labour Department
Auditor compliance certificate
Approval from Chief Electrical Inspector
Clearance from Forest department

Agreements

- Power Evacuation arrangement permission letter from Gov.
- Confirmation of Metering Arrangement and location
- Meter type, Manufacture, Model, Details for Energy Metering
- Copy of PPA (important as Preferential PPA projects are not eligible for REC)
- Proposed Model and make of plant equipment
- Taking for compliance with the usage of fossil fuel criteria as specified by MNRE
- Details of Connectivity with Gov. Grid
- Connectivity Diagram and Single Line Diagram of Plant
- Any other documents requested by State Gov.

Land purchase

3.Financial objectives

Cost objectives

Per Watt	**Rs.35 - 50**
Per MW	**Rs.3.3 - 3.7 Crore**
Per MW	**4.5 - 5.0 Acre**
Per Acre	**Rs.25-40 Lakhs**
O&M Cost	**Rs.15 Lakhs/MW/Year**
Insurance	**0.5% per year**
AD Benefit	**4.21% per year**
Escalation Charges	**2% every year**

Year	Loss	Unit Generation	EB Tariff	Amount
1	0.0066	14,40,000	3.00	43,20,000
2	0.0066	14,39,000	3.00	43,17,000
3	0.0066	14,38,000	3.00	43,14,000
4	0.0066	14,37,000	3.00	43,11,000
5	0.0066	14,35,000	3.00	43,05,000
6	0.0066	14,30,000	3.00	42,90,000
7	0.0066	14,20,000	3.00	42,60,000
8	0.0066	14,15,000	3.00	42,45,000
9	0.0066	14,10,000	3.00	42,30,000
10	0.0066	14,05,000	3.00	42,15,000
11	0.0066	14,00,000	3.00	35,43,000

Year	**20 years**
Gen	**Units**
Tariff	**2% Es**
Amt	**Rs**

12	0.0066	13,90,000	3.00	37,53,000
13	0.0066	13,00,000	3.00	39,00,000
14	0.0066	13,00,000	3.00	39,00,000
15	0.0066	13,00,000	3.00	39,00,000
16	0.0066	13,00,000	3.00	39,00,000
17	0.0066	13,00,000	3.00	39,00,000
18	0.0066	13,00,000	3.00	39,00,000
19	0.0066	13,00,000	3.00	39,00,000
20	0.0066	13,00,000	3.00	39,00,000

Year	Capital Cost -3,50,00,000	Cash Flow
1	-ve	-43,20,000
2	-ve	-43,20,000
3	-ve	-43,20,000
4	-ve	-43,20,0006

Payback Period

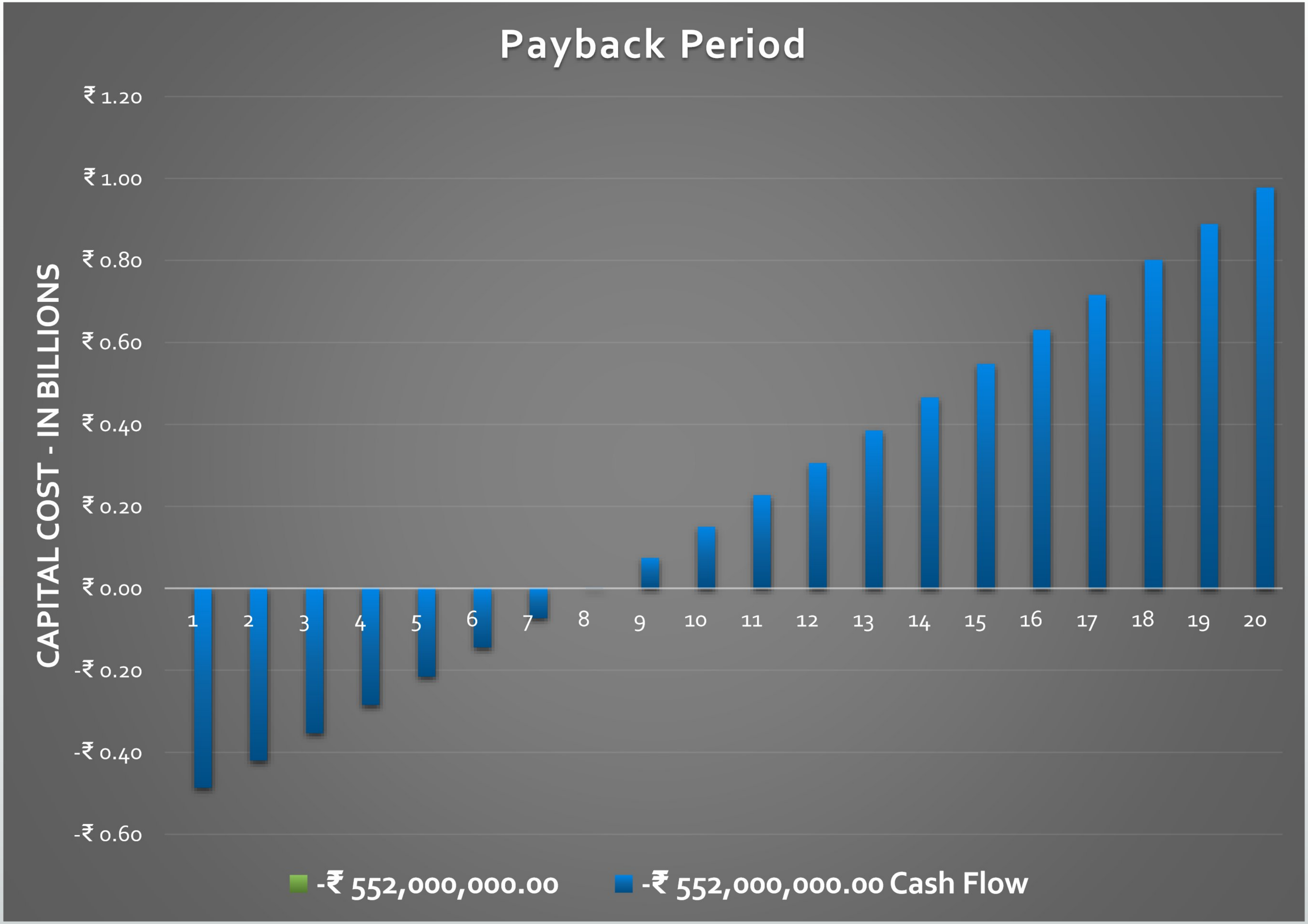
7 Years -ve

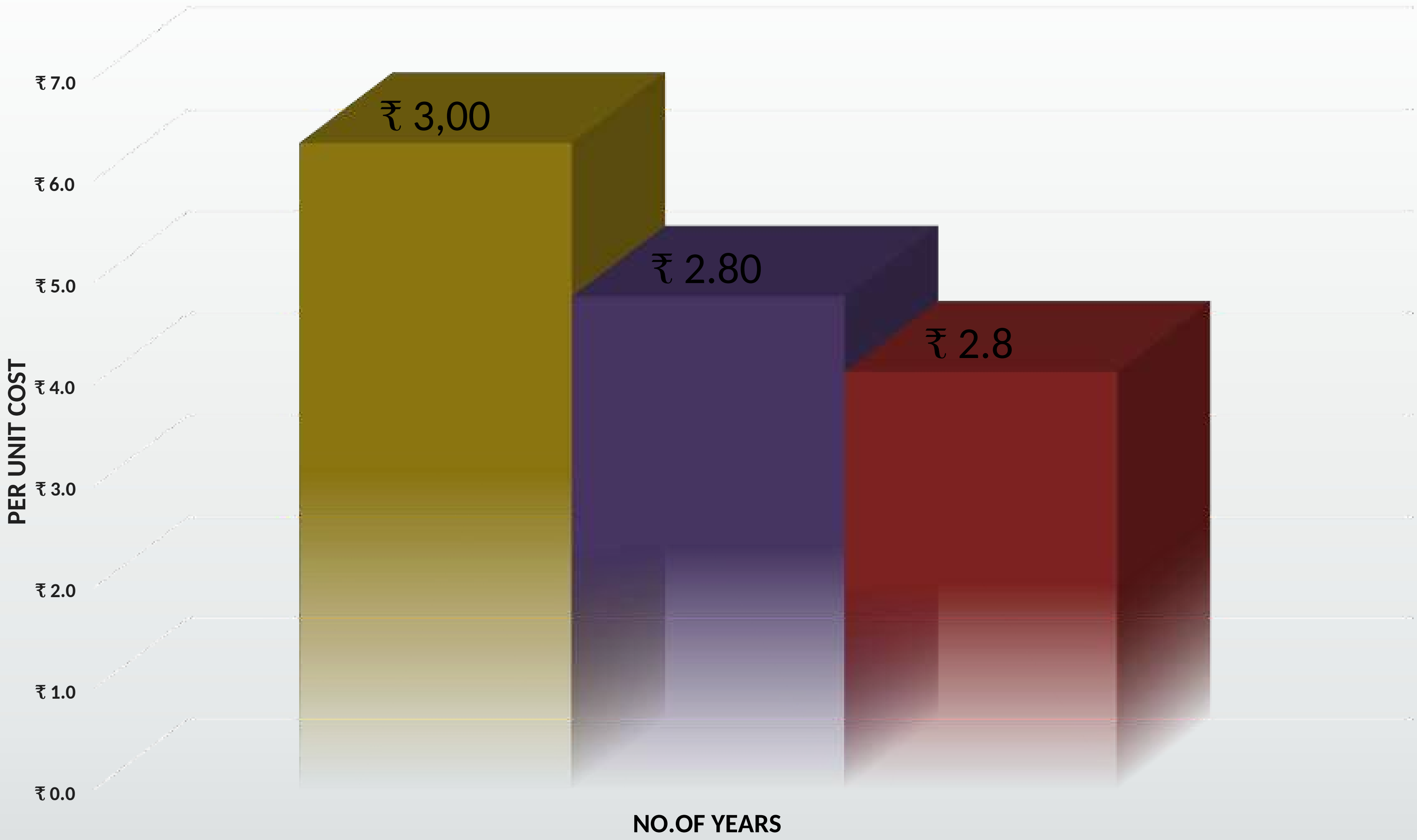
Then +ve

Net Amount

Gen.Amount -Total Expenses

5	-ve	-43,20,000
6	-ve	-43,20,00043
7	-ve	-43,20,000
8	+ve	39,00,000
9	+ve	39,00,000
10	+ve	39,00,000
11	+ve	39,00,000
12	+ve	39,00,000
13	+ve	39,00,000
14	+ve	39,00,000
15	+ve	39,00,000
16	+ve	39,00,000
17	+ve	39,00,000
18	+ve	39,00,000
19	+ve	39,00,000
20	+ve	39,00,000


Payback Period
CAPITAL COST - IN BILLIONS
₹ 1.20
₹ 1.00
₹ 0.80
₹ 0.60
₹ 0.40
₹ 0.20
₹ 0.00
-₹ 0.20
-₹ 0.40
-₹ 0.60
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
-₹ 552,000,000.00
-₹ 552,000,000.00 Cash Flow


₹ 7.0
₹ 6.0
₹ 5.0
₹ 4.0
₹ 3.0
₹ 2.0
₹ 1.0
₹ 0.0
PER UNIT COST
₹ 3,00
₹ 2.80
₹ 2.8
NO.OF YEARS

10 years 15 years 20 years

Bank loans for Solar PV Plant setup

2 kinds of Financing mechanisms

- **Recourse Financing**
- **Non-Recourse Financing**

Loan could be in the range of 40-60% project cost

CDM & REC Benefits

REC Mechanism and CDM are mutually exclusive and hence a power developer can claim CDM benefits (Carbon Credits) also.

- **REC Principal Regulations, there is no lower limit for Solar Power plants to be eligible for RECs. Though it previously stated that 250 KW is the minimum size for plants to be eligible for RECs, the same has been removed as part of the above stated amendment.**

No Subsidy

Central Government through MNRE

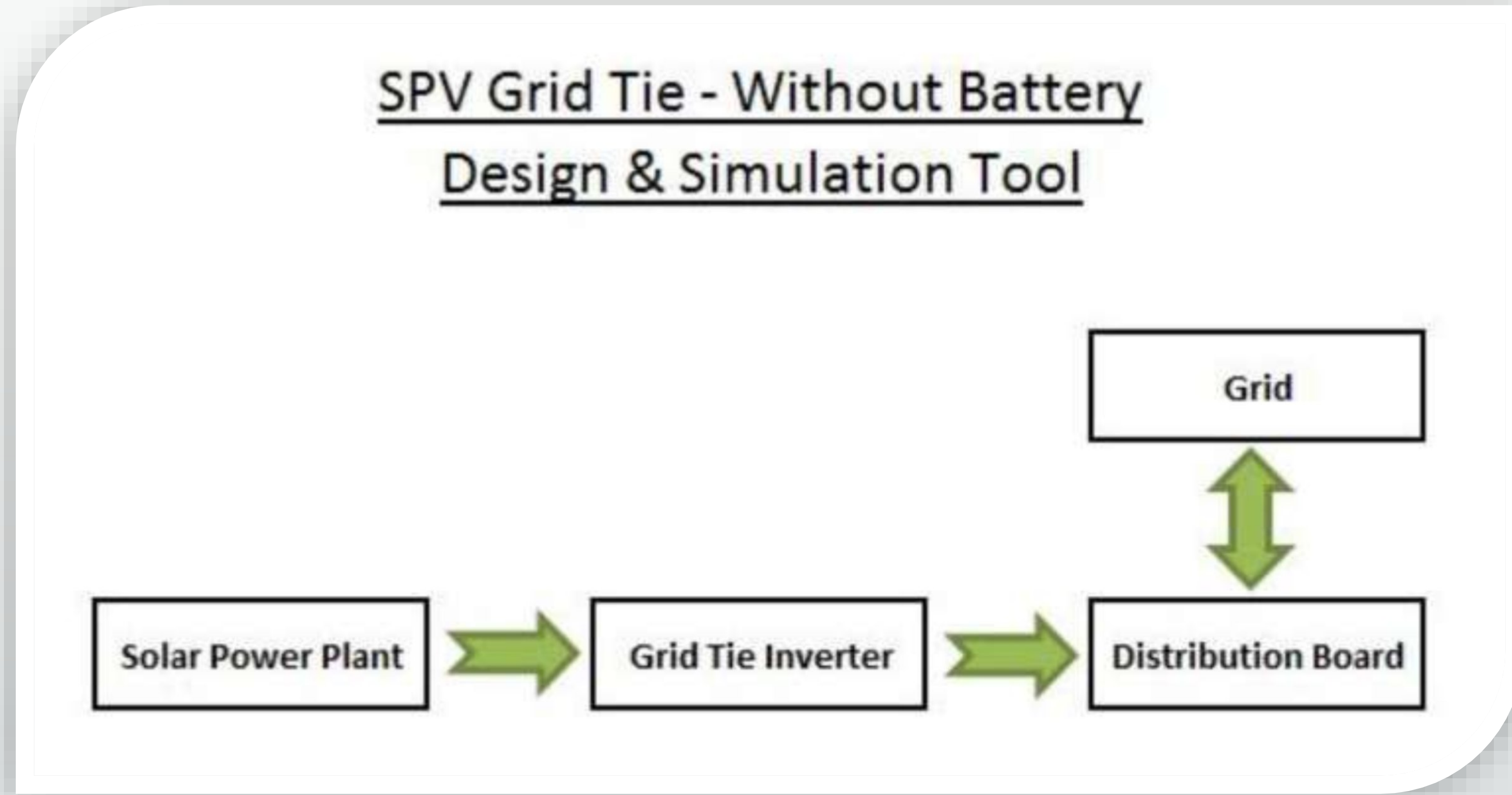
- For systems Upto 100 kWp in size, Upto 15% subsidy can be availed with the help of MNRE-empanelled channel partners.

Solar Energy Corporation of India

- For systems of sizes 100 kWp-500 kWp, subsidy can be availed through Solar Energy Corporation of India.

4.Technical objectives

On-Grid



Details

- Plant Capacity 1 MW
- Solar radiation 4-7 kWh/m^2/day
- Hrs of operation/Day 5-7 Hrs
- Plant working days 300 Days
- Ave.annual Production 12.6 M.U (10*6*300 MWhr)
- Inverter Central Inverter
- Inside Losses 5 % from generation

Also calculate the Number of sunny days, Day Temperatures, Air Mass….etc

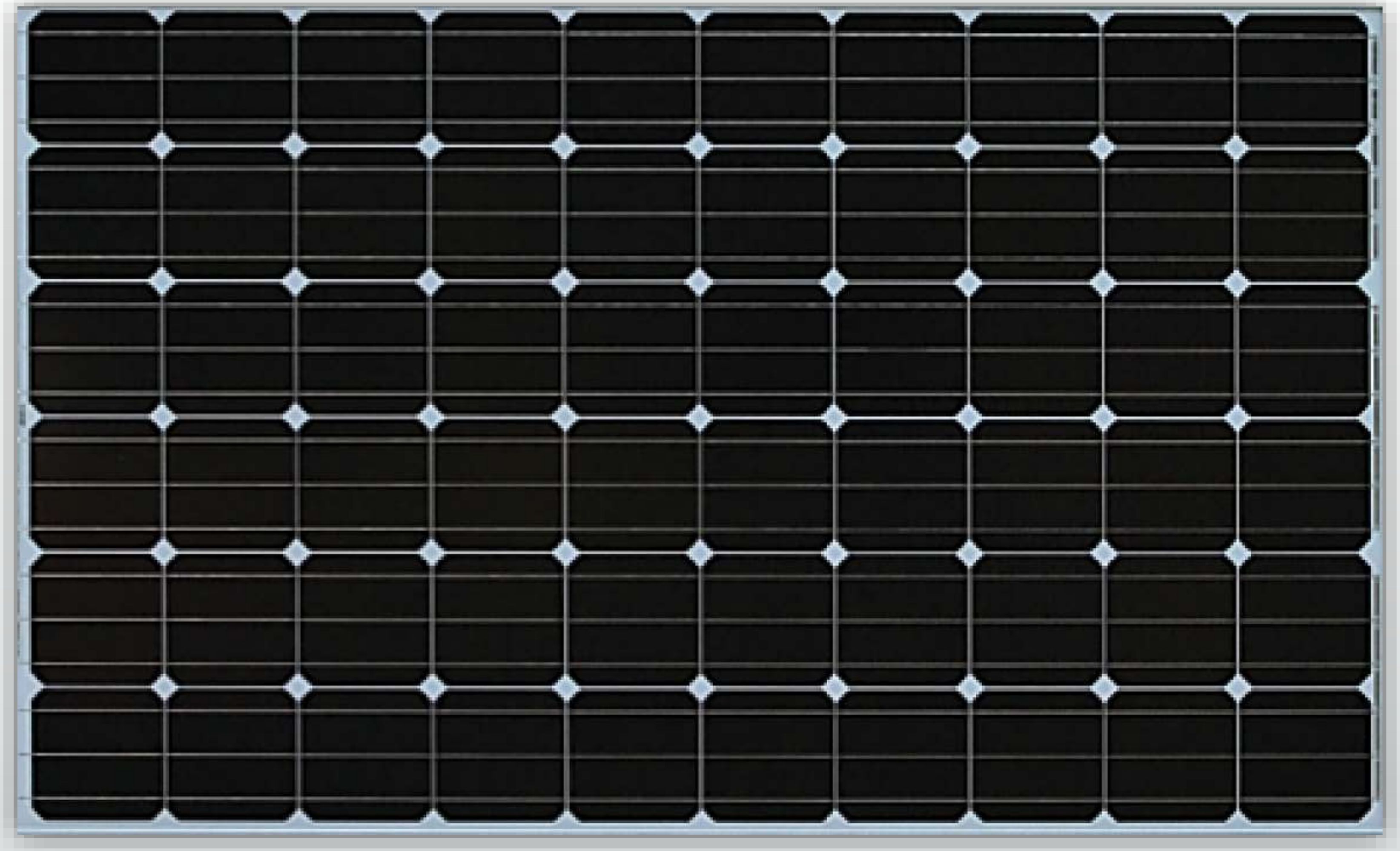
Site Assessment

Prepare Plant Layout	**Surveying & Now using Google Earth with ArcGIS**
Latitude	**9^0-31^0**
Area of plant	**No's of modules (55 Acres)**
Available shadow area	**Waste land - Ground**
Soil Type	**All Coarse / Fine Soils used**
Foundation	**100 mm thickness**

250 Wp - Panel Specifications

Solar Panel Watt Peak	**235 W**
Module Type	**Standard**
Technology	**Mono crystalline**
Mounting Disposition	**Ground - Tilted**
Open circuit voltage	**37 V**
Short circuit current	**8.6 A**
Max.Power voltage	**30 V**
Max.Power current	**7.84 A**

250 Wp - Panel



Design Calculation

Total Plant Size	-	10 MW
Individual Sections	-	500 kW * 20 Sets
No.of DC Cabinet	-	560 V * 20 Sets
No.of AC Cabinet	-	380 V * 20 Sets
No.of Inverters	-	500 kVA * 20 Sets
No.of Junction Boxes	-	560 W * 320 Sets
Monitoring Devices	-	1 Set
Protection Devices	-	1 Set
Infrastructure	-	As required

Each 500 kW Set

PV Arrays	-	16 Sets

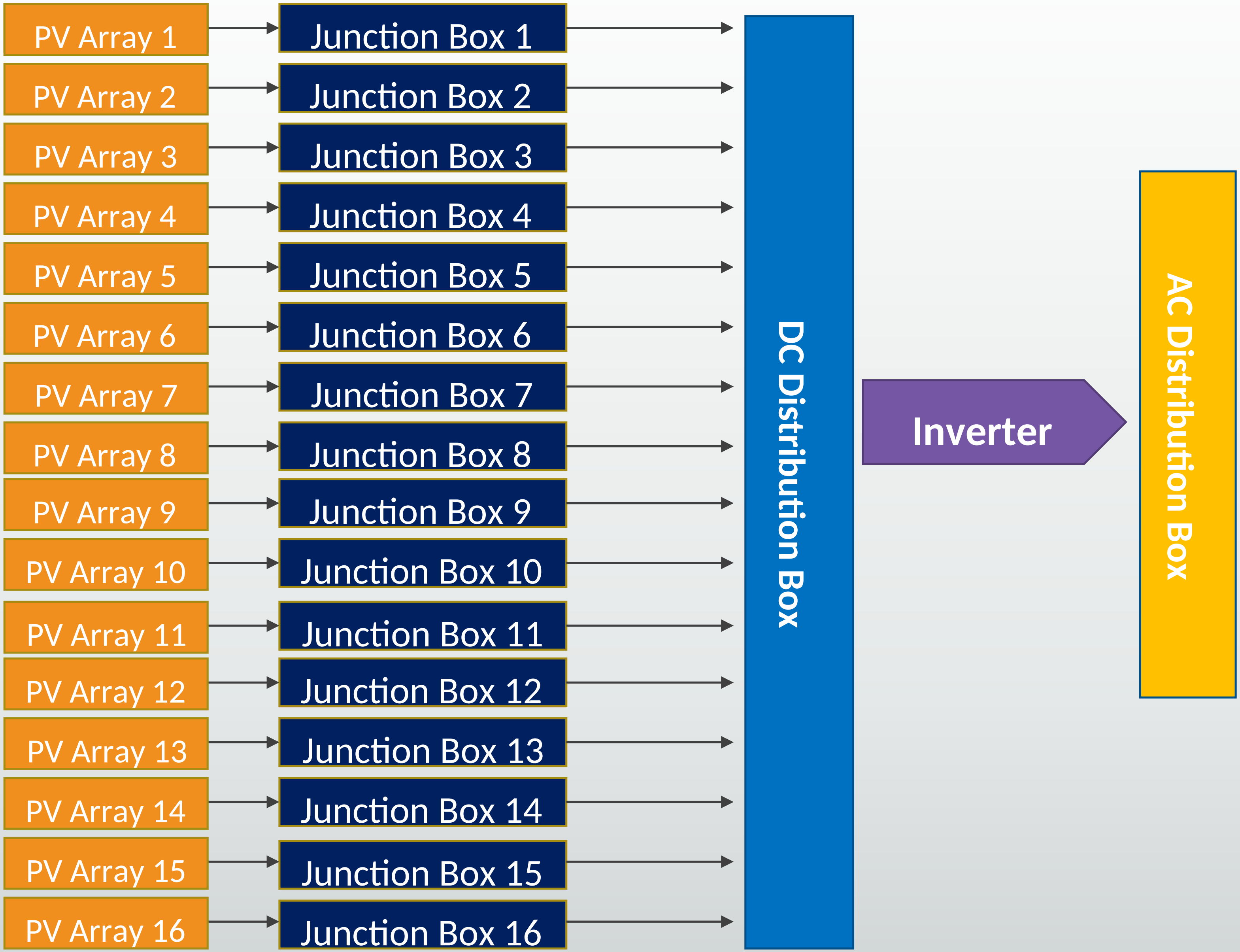
PV Array – 16

Mounting Structure	-	16 Sets
DC Junction Box (560 W)	-	16 Sets
DC Cabinet (560 V)	-	1 Set

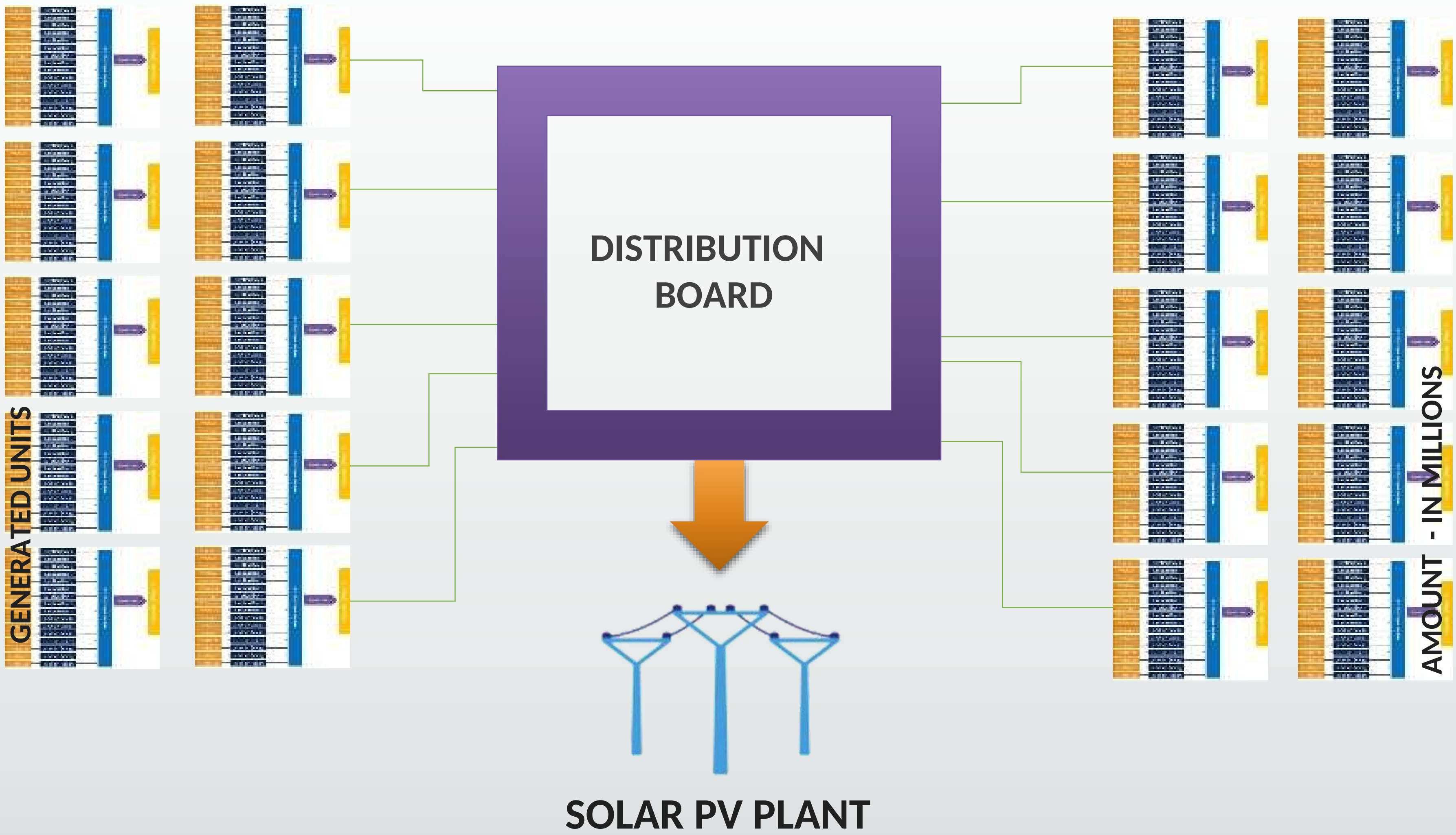
Inverter (500 kVA) - 1 Set AC Distribution Cabinet - 1 Set

15 PV Arrays	-	Has 8 Rows
Last PV Array	-	Has 5 Rows
Each Row	-	Has 16 Modules
Net Panels (500 kW)	-	(8*16) + (5*16) = 2000 Panels
Like 20 Sets (10 MW)	-	20*2000 = 40000 Panels

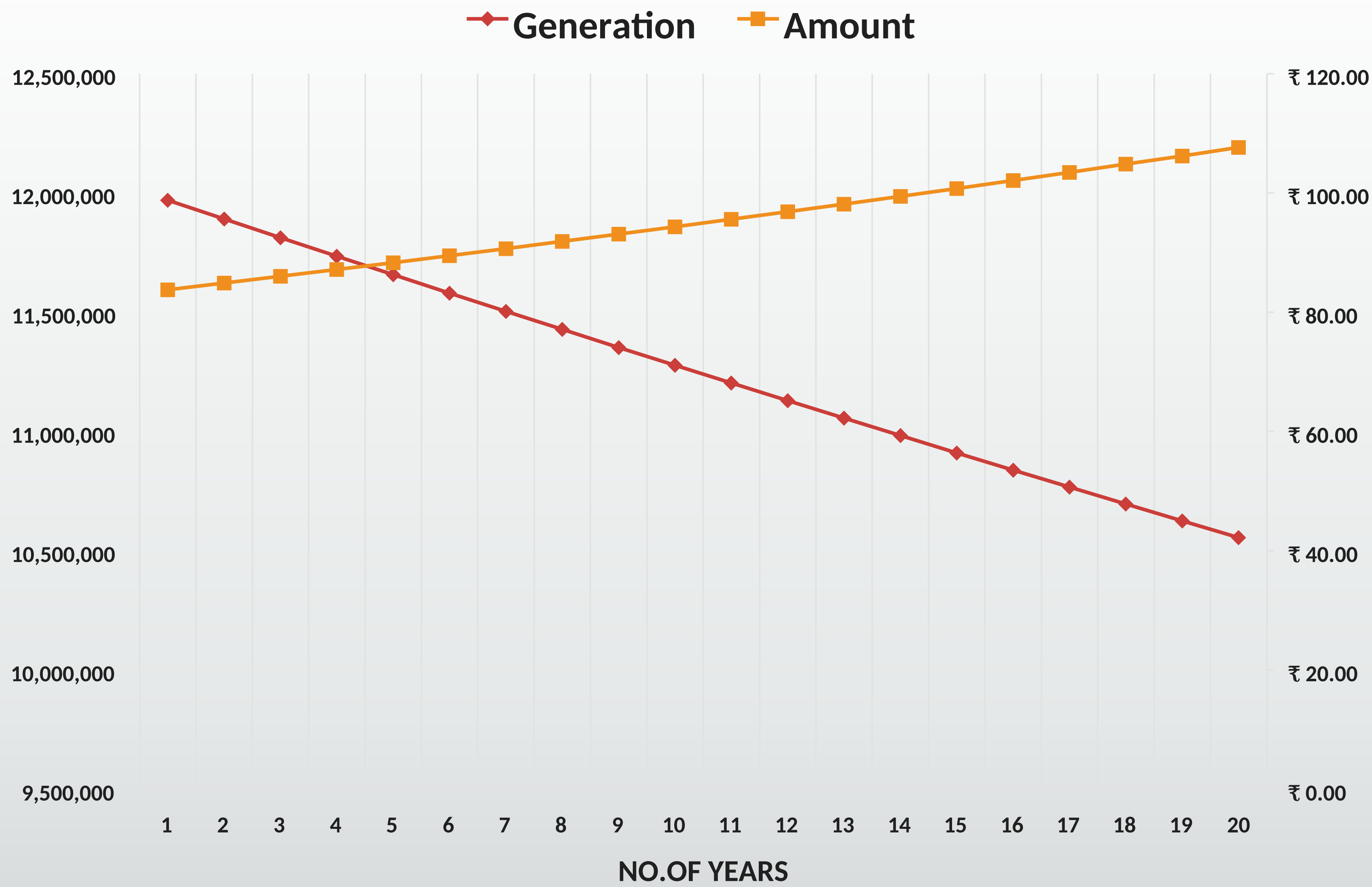
500 kW - Design layout


PV Array 1
PV Array 2
PV Array 3
PV Array 4
PV Array 5
PV Array 6
PV Array 7
PV Array 8
PV Array 9
PV Array 10
PV Array 11
PV Array 12
PV Array 13
PV Array 14
PV Array 15
PV Array 16
Junction Box 1
Junction Box 2
Junction Box 3
Junction Box 4
Junction Box 5
Junction Box 6
Junction Box 7
Junction Box 8
Junction Box 9
Junction Box 10
Junction Box 11
Junction Box 12
Junction Box 13
Junction Box 14
Junction Box 15
Junction Box 16
DC Distribution Box
Inverter
AC Distribution Box

10 MW Design layout



DISTRIBUTION BOARD
SOLAR PV PLANT
GENERATED UNITS
AMOUNT - IN MILLIONS



Generation
Amount
12,500,000
12,000,000
11,500,000
11,000,000
10,500,000
10,000,000
9,500,000
₹ 120.00
₹ 100.00
₹ 80.00
₹ 60.00
₹ 40.00
₹ 20.00
₹ 0.00
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
NO.OF YEARS

CER Calculation

Amount of ele.pro/year	12600000	Units
CER	12600	Units
1 CER	1	Ton of Co2
1 CER	0.4	Euro
1 Euro	72	Rupees
Total CER (Euro)	*5040*	*Euro*

Total Value/Annum	***₹ 362,880.00***	***Rupees***
For 15 Years	***₹ 5,443,200.00***	***Rupees***

5.Special objectives

Success Factors

For Output

- Plant Location
- Quality of equipment used
- Solar Tracking systems
- O&M activities

For Project

- Satisfied clients
- Met project objectives
- Completed within budget
- Delivered on time

Our Project Pathway

Proposal

- November 6

Permission

- November 13

Financial

- November 20

Report

- Today

Appendix

- Ministry of New & Renewable Energy (MNRE) Website
- Solar PV Systems – JRR Energies